

CIA TRIAD ANALYSIS REPORT

Cybersecurity Internship – Task 2

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Course:

B.Voc Cybersecurity and Forensics

Task:

Task 2 – CIA Triad Analysis

Track:

Cyber Security

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Introduction

Cybersecurity is the practice of protecting computer systems, networks, applications and data from unauthorized access, modification, destruction and disruption.

The CIA Triad is one of the fundamental concepts of information security. CIA stands for **Confidentiality, Integrity and Availability**.

These three principles help organizations protect information and maintain secure and reliable systems. A cybersecurity incident can affect one or more of these principles.

The CIA Triad is therefore used to understand security risks and analyze the impact of cyberattacks.

What is the CIA Triad?

The CIA Triad is a security model consisting of three important principles:

- **Confidentiality**
- **Integrity**
- **Availability**

1. Confidentiality

Confidentiality means protecting information from unauthorized access or disclosure. Only authorized users should be able to access sensitive information.

Examples of confidential information

- Passwords
- Bank account details
- Credit card information
- Personal information
- Medical records
- Business secrets
- Government information

Methods used to protect confidentiality

- Passwords
- Multi-factor authentication
- Encryption
- Access control
- User permissions
- Data classification

Example

If a hacker steals a company's customer database containing names, addresses and financial information, confidentiality has been violated because unauthorized people obtained private information.

2. Integrity

Integrity means ensuring that information remains accurate, complete and trustworthy. Data should not be changed, deleted or manipulated by unauthorized people.

Examples

- Bank balance should not be changed illegally.
- Student marks should not be modified without authorization.
- Company records should remain accurate.
- Software should not contain unauthorized malicious modifications.

Methods used to protect integrity

- Hashing
- Digital signatures
- Access control
- File permissions
- Backups
- Version control
- Audit logs

Example

If an attacker changes a student's examination marks in a database without authorization, the integrity of the data has been compromised.

3.Availability

Availability means that authorized users should be able to access systems, applications and information whenever they need them.A system that is secure but unavailable cannot effectively provide its services.

Threats to availability

- **DDoS attacks**
- **Ransomware**
- **Hardware failure**
- **Network failure**
- **Power failure**
- **Natural disasters**
- **Server crashes**

Methods used to maintain availability

- **Regular backups**
- **Disaster recovery**
- **Redundant servers**
- **Load balancing**
- **Backup power**
- **Network monitoring**
- **Incident response plans**

Example

If a banking website becomes unavailable because of a DDoS attack, customers may be unable to access their accounts. This is an availability failure.

Five Real-World Examples

Example 1 — Equifax Data Breach

- **Equifax Data Breach – Confidentiality Failure**

Incident

In 2017, Equifax experienced a major cybersecurity breach in which attackers gained access to sensitive personal information.

According to the U.S. Federal Trade Commission, approximately **147 million people** were affected. The information included names, dates of birth, Social Security numbers, addresses and other sensitive information.

CIA Classification

>>Confidentiality

Why?

The attackers obtained sensitive information that should have been protected from unauthorized access.

Therefore, the incident represents a major **confidentiality failure**.

Security Lessons

Organizations should:

- Patch vulnerabilities quickly.
- Use strong access controls.
- Encrypt sensitive information.
- Monitor networks for suspicious activity.

- Avoid storing credentials in plain text.
- Regularly perform security assessments.

Impact

The exposure of personal information created risks such as identity theft and fraud for affected individuals.

Example 2 — SolarWinds

- **SolarWinds Supply-Chain Attack – Integrity Failure**

Incident

In 2020, attackers compromised the SolarWinds Orion software supply chain. Malicious code was inserted into affected versions of the software, which were subsequently distributed to customers.

CISA reported that the incident involved compromise of SolarWinds Orion software and affected government agencies, critical infrastructure organizations and private-sector organizations.

CIA Classification

>>Integrity

Why?

Software that customers trusted had been modified by an unauthorized party.

The attack therefore demonstrates a failure of **integrity**, because trusted software was altered.

Security Lessons

Organizations should:

- Secure the software development process.
- Verify software updates.
- Use digital signatures.
- Monitor supply-chain risks.
- Apply network segmentation.
- Monitor unusual software behavior.

Impact

The incident demonstrated how compromising a trusted software supplier can allow attackers to affect multiple organizations.

Example 3 — WannaCry and NHS

- **WannaCry Ransomware Attack – Availability Failure**

Incident

In May 2017, the WannaCry ransomware attack affected organizations around the world, including the UK's National Health Service.

The UK's National Audit Office reported that at least **81 out of 236 NHS trusts** in England were affected, along with hundreds of other healthcare organizations. The attack disrupted healthcare services.

CIA Classification

>>Availability

Why?

Ransomware encrypted files and prevented organizations from accessing systems and information.

Therefore, authorized users could not access required systems normally.

This represents an **availability failure**.

Security Lessons

Organizations should:

- Regularly patch systems.
- Replace unsupported operating systems.
- Maintain offline backups.
- Use endpoint protection.
- Segment networks.
- Have an incident response plan.

Impact

The attack disrupted healthcare services and affected the ability of NHS organizations to provide normal services.

Example 4 — Colonial Pipeline

- **Colonial Pipeline Ransomware Attack – Availability Failure**

Incident

In May 2021, a ransomware attack affected the IT network of Colonial Pipeline, a major U.S. energy infrastructure organization.

CISA and the FBI described the incident as a ransomware attack against a critical infrastructure organization and issued recommendations aimed at reducing business disruption from ransomware.

CIA Classification

>>Availability

Why?

The attack resulted in a suspension of operations. When important systems are unavailable, normal business operations can be disrupted.

Therefore, the incident demonstrates an **availability failure**.

Security Lessons

Organizations should:

- Maintain tested backups.
- Separate IT and operational technology networks.
- Use network segmentation.
- Monitor systems.
- Prepare incident response plans.
- Test recovery procedures.

Impact

The incident demonstrated how a cyberattack against critical infrastructure can cause significant operational disruption.

Example 5 — GitHub 2018 Incident

- **GitHub 2018 Incident – Availability and Integrity**

Incident

In October 2018, GitHub experienced an incident involving a network partition and database problems.

GitHub reported that multiple services were affected, information displayed on the website became outdated or inconsistent, and several services were unavailable or degraded for approximately 24 hours. GitHub also reported that no user data was lost.

CIA Classification

>>Availability and Integrity

Why?

Availability:

Some GitHub services could not operate normally.

Integrity:

Some information displayed on the website became outdated or inconsistent.

GitHub prioritized protecting data integrity while restoring its services.

Security Lessons

Organizations should:

- Use reliable database replication.
- Maintain redundant systems.
- Monitor system health.
- Have disaster recovery procedures.
- Test failover systems.
- Maintain accurate incident response procedures.

● **How Organizations Can Protect the CIA Triad**

Organizations should implement multiple layers of cybersecurity controls.

For Confidentiality

- Encryption
- Strong passwords
- Multi-factor authentication
- Access control
- Least privilege
- Data classification

For Integrity

- Hashing
- Digital signatures
- File integrity monitoring
- Audit logs
- Version control
- Secure software development

For Availability

- Backups
- Redundant systems
- Disaster recovery
- DDoS protection
- Network monitoring
- Incident response plans

Conclusion

The CIA Triad is a fundamental concept in cybersecurity. Confidentiality protects information from unauthorized access, integrity protects information from unauthorized modification, and availability ensures that authorized users can access systems and information when required.

The real-world incidents discussed in this report demonstrate that cybersecurity failures can affect organizations and users in different ways. A single incident may affect more than one CIA principle.

References

You can put these at the end of your report:

1. U.S. Federal Trade Commission – Equifax Data Breach
2. U.S. Cybersecurity and Infrastructure Security Agency (CISA) – SolarWinds Software Compromise
3. UK National Audit Office – WannaCry Cyber Attack and the NHS
4. CISA/FBI – Ransomware and Colonial Pipeline
5. GitHub Blog – October 2018 Incident Report

These are reliable sources because they are official government/company sources.

Interview Questions

Q1. What is Confidentiality?

Answer:

Confidentiality means protecting information from unauthorized access or disclosure. Only authorized users should be able to access sensitive information.

Q2. What is Integrity?

Answer:

Integrity means maintaining the accuracy, completeness and trustworthiness of information and preventing unauthorized modification.

Q3. What is Availability?

Answer:

Availability means ensuring that authorized users can access systems, applications and information whenever they are required.

